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Business Proposal

UMHackathon 2026

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1. Executive Summary

AdKedai is a web-based, AI-powered inventory-based advertising decision and content generation platform built specifically for small and medium-sized e-commerce sellers in Kuala Lumpur. The platform solves a critical and persistent gap: KL SME sellers operating on Shopee, Lazada, and TikTok Shop have products in stock but no structured, data-driven guidance on how to advertise them effectively. Most rely on guesswork such as choosing platforms, formats, and posting times without any supporting data. According to the e-Conomy SEA 2023 report by Google, Temasek, and Bain & Company, the SEA e-commerce market reached USD 159 billion in gross merchandise value in 2023 and is projected to grow to USD 295 billion by 2028. Despite this enormous opportunity, over 65% of SME sellers make advertising decisions based on intuition or peer advice rather than data (Lazada Internal Seller Research, 2022), resulting in an estimated 37–42% waste of digital ad spend (Meta Business, 2023).

AdKedai addresses this by leveraging Z.AI's GLM as its sole intelligence engine. Sellers upload their CSV stock data, and AdKedai's four-stage agentic pipeline classifies inventory, generates ranked advertising strategy recommendations contextualized against KL market conditions and SEA seasonal events, and produces complete text and static image ad content package, without requiring any marketing expertise.

AdKedai's revenue model operates on two tracks.

- Primary track is a B2B SaaS Tier plan at RM 1,500 per month, designed for KL-based marketing agencies and businesses managing multiple SME seller clients under a single dashboard.
- Secondary track is open to the public through three self-serve tiers: Free at RM 0 with 1 session per month, Starter at RM 49 per month with 6 sessions per month including text and image generation, and Pro at RM 149 per month with unlimited sessions, text and image generation, and full session history.

AdKedai estimates approximately RM 1.0 million in Year 1 revenue from its Kuala Lumpur beachhead market, growing to an estimated RM 7.2 million by Year 2 as the platform expands Malaysia-wide, and reaching an estimated RM 28.5 million by Year 3 upon initial SEA market entry. Based on these projections, AdKedai estimates achieving profitability between Month 28 and Month 30, with a net positive position of approximately RM 10.5 million by end of Year 3.

2. Product Overview / Solution

- **Inventory Intelligence via Data Upload:** Sellers upload a CSV file containing product name, category, stock level, price, date added and expired date. AdKedai's pipeline which powered by Z.AI's GLM, classifies each product by promotional urgency, overstock status, and seasonal relevance to upcoming SEA e-commerce sales events such as 11.11, 12.12, Hari Raya, and Chinese New Year.
- **AI Ad Strategy Recommendations:** GLM cross-references inventory data with real-time SEA market trends and the seasonal event calendar to generate at least 2 ranked advertising strategy recommendations per session. Each recommendation covers image generation, captions, ad copy, and hashtags, supported by plain-language explanations for platform choice, target audience, optimal timing, and geographic targeting. It also includes estimated performance metrics (reach, clicks, sales, revenue, and ROI) calculated using market benchmarks and product data.
- **Seller Review & Decision Dashboard:** All recommendations are displayed as interactive cards in a clean dashboard. Sellers review AI reasoning, compare options, and select their preferred strategy before any content is generated, preserving full human oversight of every advertising decision.
- **Ad Content Refinement via Prompt:** User can select any specific part they wish to refine. By sending a free-text prompt describing the desired change, such as requesting more hashtags, a longer headline, a more casual caption tone, or a more urgent call to action, GLM regenerates only the selected element while keeping the rest of the content package unchanged. Ensuring the published ad content accurately reflects their intent before exporting.
- **Ad Content Generation:** Based on the selected strategy, CSV product details and editing, GLM generates a complete, ready-to-publish ad content package: headline, body copy, platform-optimized caption, curated hashtag set, and posting guidance, validated post-generation to ensure content accurately reflects product name, category, and pricing.
- **Session History:** All generated advertising strategies and ad content packages are automatically saved to the seller's account history after each session. Sellers can revisit, review, and reference any previously generated strategy recommendations and ad content at any time through their

personalized dashboard, without needing to re-upload their inventory or regenerate content from scratch.

- **Personal Account System:** Sellers register, log in, and access their full recommendation and generated content history from a personalized dashboard at any time.

Technology Used:

- **React (Frontend) + Python / FastAPI / Uvicorn (Backend):** A cloud-server architecture with modular microservices. The React frontend handles login, CSV upload, dashboard display, ad previews, and exports. FastAPI manages API requests, CSV validation, AI workflow orchestration, and session caching.
- **Z.AI GLM-5.1 (Core AI Intelligence Engine):** Powers all text-based pipeline stages — inventory classification, strategy recommendation generation, ad content generation, and reasoning explanation via structured JSON prompting with schema validation and up to two automatic retries per stage on failure.
- **Z.AI GLM-Image (Visual Content Generation):** Generates platform-ready product ad images from structured prompts derived from the seller's CSV data and selected advertising strategy.
- **Firebase Firestore (Database & Authentication):** Stores user accounts, session history, uploaded CSV records, recommendation outputs, and generated content. Supports real-time data management and authentication services.
- **Vercel (Frontend) + AWS (Backend) Deployment:** Vercel provides global CDN delivery of the React frontend. AWS hosts the FastAPI backend with async request handling, ensuring scalability during peak e-commerce periods.
- **GitHub Actions CI/CD Pipeline:** Automated build, unit test, linting, and regression test gates enforce quality thresholds, minimum 90% test coverage, zero P0/P1 bugs, $\geq 80\%$ AI output pass rate, before any deployment to production.

3. Market Research & Market Dynamics

3.1 Problem Landscape

E-commerce sellers in Southeast Asia face a persistent structural gap between having products to sell and knowing how to advertise them effectively. Existing platform tools such as Shopee Ads and TikTok for Business provide execution infrastructure for running ads but offer no strategic guidance. These platform assume the seller already has a strategy, and that is exactly the gap AdKedai is built to fill.

A 2022 study by Lazada's internal seller research division found that over 65% of SME sellers made advertising decisions based on intuition or peer advice rather than data, resulting in wasted budgets, unsold inventory, and missed seasonal sales opportunities. Meta Business (2023) estimated that SMEs in SEA waste approximately 37–42% of their digital ad spend due to poor targeting and timing decisions. Hiring a marketing professional in Malaysia costs between RM 3,000 and RM 8,000 per month (MDEC, 2023), placing professional advertising strategy beyond the reach of most small sellers. The Journal of Small Business Management (2022) found that SME and e-commerce owners spend an average of 6.3 hours per week on marketing-related tasks, with ad planning and content creation accounting for the majority of that time.

3.2 Market Niche

AdKedai primarily targets digitally active SME sellers operating independently on Shopee, Lazada, and TikTok Shop within Kuala Lumpur — particularly those managing stores without a dedicated marketing team or formal advertising knowledge. This includes fashion sellers, beauty merchants, FMCG resellers, and electronics traders who experience high inventory turnover and strong sensitivity to seasonal sales events. Secondary targets include KL-based SME-focused marketing agencies seeking to scale campaign output across multiple seller clients. Kuala Lumpur serves as AdKedai's beachhead market, with structured expansion to Selangor, Penang, and Johor by Year 2, and broader SEA coverage by Year 3.

3.3 Market Sizing

Market Tier	Definition	Estimated Value (2026)
TAM (Total Addressable Market)	AI-assisted advertising tools for all SME e-commerce sellers across Southeast Asia (Malaysia, Indonesia, Thailand, Philippines, Vietnam & Singapore)	RM 2.1 billion*
SAM (Serviceable Addressable Market)	Digitally active SME sellers on Shopee, Lazada & TikTok Shop within Malaysia who actively spend on digital advertising	RM 68 million*
SOM (Serviceable Obtainable Market)	Realistic market capture within Kuala Lumpur, AdKedais beachhead market for the 18-month launch horizon (~1,700–2,500 paying subscribers)	RM 3.2 million*

*Market sizing figures are internal estimates derived from publicly available data sources including the e-Conomy SEA 2023 report (Google, Temasek & Bain), MDEC 2023 SME Digitalisation Data, and DataReportal Malaysia 2024. These figures are projections and not sourced from a single published market research report. Actual market size may vary based on market conditions.

3.4 Market Dynamics & Placement:

There are four key dynamics that shape AdKedai's entry into this market:

- **E-Commerce Growth Momentum:** The SEA e-commerce market is projected to grow from USD 159 billion in 2023 to USD 295 billion by 2028 (Google, Temasek & Bain, 2023). Accelerating seller competition on Shopee, Lazada, and TikTok Shop increases the urgency for data-driven advertising differentiation among KL SME sellers.
- **Platform Fragmentation Across Channels:** KL sellers must manage advertising across Shopee, Lazada, TikTok Shop, Instagram, and Facebook simultaneously. Each with different ad formats, audiences, and posting norms. No existing tool unifies inventory-based strategy recommendations across all these platforms in a single workflow designed for SME sellers.



- **AI Adoption Among SMEs:** MDEC's 2024 Digital Economy Blueprint identifies AI adoption as a national priority for Malaysian SMEs, with government grant programmes subsidising AI productivity tool adoption. This policy tailwind reduces AdKedai's customer acquisition cost and legitimises AI-assisted advertising for first-time adopters in KL.
- **Seasonal Revenue Concentration:** KL e-commerce revenue is highly concentrated around seasonal sales events such as 11.11, 12.12, Hari Raya, Chinese New Year, and Valentine's Day. AdKedai's seasonal recommendation engine directly addresses this by proactively identifying which products in a seller's inventory are most relevant for each upcoming event.

4. Value Proposition & Competitive Advantage

AdKedai redefines the economics and accessibility of e-commerce advertising strategy for KL SME sellers. Where existing platforms tell sellers how to run ads, AdKedai tells them what to advertise, where, when, and to whom, helps derived directly from their own uploaded stock data. The platform's core competitive advantage rests on five pillars:

- **Inventory-Grounded Intelligence:** Every strategy AdKedai generates is anchored directly to the seller's actual inventory data (product name, category, stock level, price, date added and expired date). This is fundamentally different from generic AI writing tools or platform ad managers, which have no visibility into what the seller actually has available to promote.
- **Speed from Inventory to Strategy:** AdKedai compresses what would take a marketing professional 6+ hours of manual research into under 10 minutes of automated AI analysis. This speed advantage is particularly critical during high-competition seasonal windows where sellers who act early capture significantly higher returns.
- **Accessible Price Point Versus Human Alternatives:** Professional marketing strategy in Malaysia costs between RM 3,000 and RM 8,000 per month (MDEC, 2023). AdKedai delivers comparable strategic intelligence starting from RM 49 per month for individual sellers and RM 1,500 per month for agencies managing multiple accounts. A reduction of 90–99% in cost compared to hiring a dedicated marketing professional, making data-driven advertising strategy accessible to any KL SME seller regardless of budget.
- **Human-in-the-Loop by Design:** AdKedai never auto-executes advertising decisions. Sellers review and select their preferred strategy before any content is generated and can freely refine any specific content element such as hashtags, headline, or caption tone via free-text prompt after generation. Only when the seller is fully satisfied is the final content confirmed and exported, ensuring full ownership and transparency at every step.
- **Multi-Platform, SEA-Contextualised Recommendations:** Unlike platform-specific tools, AdKedai generates strategy recommendations across Shopee, Lazada, TikTok Shop, Instagram, and Facebook simultaneously, which enabling sellers to allocate their advertising budget across platforms based on data rather than familiarity.

5. Competitor Analysis (Porter's Five Forces)

Force	Intensity	Assessment
Competitive Rivalry	Medium	No direct competitor currently combines stock data analysis, strategy recommendation, and ad content generation in a single AI pipeline for KL SME sellers. Existing tools such as Shopee Ads, TikTok for Business, Graas.ai, AdStellar, and Mintly each address only part of the workflow, none close the complete stock-to-ad gap that AdKedai fills.
Threat of New Entrants	Medium	Building a reliable multi-stage agentic AI pipeline with accurate KL and SEA-contextualised recommendations requires significant technical investment and prompt engineering expertise. However, large platforms such as Shopee and TikTok with existing seller data and AI resources could develop in-house equivalents within a 12–18 month horizon, representing the most credible long-term competitive threat.
Bargaining Power of Buyers	High	KL SME sellers are highly price-sensitive and face low switching costs. AdKedai mitigates this through session history lock-in, clear time-saving value, and a tiered pricing model (Free Trial, RM49 Starter, RM149 Pro) that lowers entry barriers while encouraging long-term retention.
Bargaining Power of Suppliers	High	AdKedai currently relies heavily on a primary LLM provider (Z.AI GLM API), creating significant supplier concentration risk. Changes in pricing, performance, or availability could directly impact core functionality. This risk is being mitigated through planned multi-provider support, along with system safeguards such as asynchronous processing, CSV caching, and graceful degradation.
Threat of Substitutes	Low	Manual ad planning, generic AI chatbots, or marketing freelancers are substitutes, but none deliver AdKedai's combination of inventory integration, SEA seasonal

		context, multi-platform strategy, and content generation in a single automated workflow.
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Competitor Comparison Table:

Competitor	Strength	Weakness	AdKedai's Advantage
Ad Execution Platforms (Shopee Ads / Lazada Ads/ TikTok for Business)	Direct access to high-intent (marketplaces) and high-discovery (TikTok) traffic; strong native ad tools; widely used by KL SMEs	Execution-focused assume sellers already know what products to promote and how to position them; no stock data integration; no decision support; siloed by platform	AdKedai acts as the decision layer across all channels by analyzing inventory to recommend what to promote, why it matters, and generating platform-adapted content for execution across Shopee, Lazada, TikTok, Instagram, and Facebook
Graas.ai	Cross-platform analytics dashboard connecting Shopee, Lazada, TikTok, Meta; good for performance reporting	Analyzes seller data but does not recommend what to advertise next. No stock data integration or content generation. Priced for larger sellers rather than KL SMEs.	Where Graas shows you the data, AdKedai takes stock data and tells you exactly what to do then generates the ad to do it.
AdStellar / Mintly	Fast AI ad creative generation; supports static images and video; covers Meta and TikTok formats	Seller must already have a strategy before using them. No stock data input, no SEA seasonal context, not designed for KL SME sellers	AdKedai helps decides <i>which product</i> to advertise based on uploaded stock data, <i>then</i> generates the content. AdStellar/Mintly require the seller to bring their own strategy.

Generic AI Tools (ChatGPT, Gemini, Jasper)	Widely accessible; flexible; low cost; can generate ad copy if prompted correctly	No stock data input; seller must manually describe their products and craft their own prompts; no KL or SEA seasonal context; no structured pipeline; results are inconsistent without prompt engineering expertise	AdKedai requires zero prompt engineering from the seller. Upload your stock CSV, select a strategy, get complete ad content. All grounded in your actual inventory and KL market context.
Marketing Freelancers / Agencies (KL)	High customisation; local KL market knowledge; relationship-based service	RM 3,000–8,000/month (MDEC, 2023); slow turnaround (days, not seconds); not scalable for seasonal spikes; completely inaccessible for most KL SME sellers on tight budgets	AdKedai delivers equivalent strategic output in under 60 seconds at RM 49–149/month and a 90–99% cost reduction with no human availability constraints, available 24/7 during peak sales events.

6. Business Model Canvas

6.1. Customer Segmentation

- **SME E-Commerce Sellers:** Independent sellers on Shopee, Lazada, and TikTok Shop across Malaysia and SEA, particularly those managing stores without a dedicated marketing team. They need fast, inventory-driven advertising decisions to compete with larger brands during peak seasonal windows.
- **SME-Focused Marketing Agencies:** Boutique agencies managing advertising campaigns across multiple SME seller clients simultaneously. AdKedai enables agencies to generate strategy and ad content at scale, reducing manual workload and accelerating campaign turnaround.
- **First-Time Entrepreneurs / New Market Entrants:** Sellers launch their first e-commerce store in KL who lack formal marketing knowledge and need structured, data-driven advertising guidance to avoid budget waste during critical launch windows.

6.2. Revenue Stream

Primary (SaaS Tier Plan):

Tier	Monthly Subscription	Target User	Number of Account	Remarks
SaaS Tier Plan	RM 1500	KL-based marketing agencies and businesses managing advertising campaigns across multiple SME seller clients simultaneously	50	Unlimited sessions per account. Each account includes full text ad content generation, static image generation, and independent session history.

Secondary (Public):

Tier	Monthly Subscription	Target User	Number of Account	Remarks
Free Trial	RM 0	First-time users, new KL SME sellers who have never used an AI advertising tool before, skeptical sellers who want to try before committing	1	1 session for one month only, full experience, text content, no image generation.
Starter	RM 49 / month	Individual or Small KL SME sellers running 1 store on Shopee, Lazada or TikTok Shop, sellers who advertise regularly but have a small product catalogue	1	6 sessions/month, , text content and image generation and session history.
Pro	RM 149 / month	Active SME sellers managing larger inventories, sellers who advertise frequently across multiple platforms, sellers who want image generation included	1	Unlimited sessions, text and image generation and session history.

6.3. Cost Structure

Cost Drivers	Cost Type
Platform maintenance, cloud infrastructure (AWS backend, Vercel frontend), Firebase Firestore operations, and ongoing product development	Fixed Cost
Z.AI GLM API token consumption per session (estimated 6,000–15,000 tokens/session; significantly reduced by CSV caching mechanism)	Variable Cost (API Usage)

Engineering and AI/prompt engineering team salaries for backend, frontend, and model evaluation	Fixed Cost (Personnel)
Customer acquisition: digital marketing, SEA seller community partnerships, agency onboarding promotions	Variable Cost (Marketing)
Ongoing prompt engineering R&D, QA testing cycles, model evaluation, and seasonal calendar maintenance	Overhead Cost (R&D)

6.4. Key Partners

Internal	External
Z.AI / ILMU (GLM-5.1 & GLM-Image)	SME Seller Communities (Facebook Groups, Shopee Seller Forums)
Firebase (Google)	MDEC & SME Corp Malaysia
AWS	Shopee & Lazada Affiliate Programmes
Vercel	Marketing Agencies (KL-based SME-focused)

6.5 Risk Assessment

Risk Type	Impact	Likelihood	Mitigation
Z.AI GLM API downtime or rate limit breach during peak seasonal periods (e.g. 11.11, Hari Raya)	High	Medium	Async session queue processes requests in order to prevent simultaneous API saturation; graceful error state and service unavailability message shown to sellers if API is unreachable
AI hallucinated or	High	Medium	JSON schema validation applied to all Stage 1

schema-invalid content reaching the seller interface			and Stage 2 GLM outputs with up to 2 automatic retries per stage on failure; post-generation content check confirms ad copy matches product name, price, and category from CSV; mandatory seller review and content refinement before final export
Invalid or low-quality CSV uploads degrading recommendation accuracy	High	Medium	Pre-GLM validation layer catches malformed files before any API call is made; files exceeding 500 rows are rejected; CSV caching serves repeated uploads without redundant GLM calls; GLM instructed to flag low-confidence classifications rather than fabricate missing product context
KL SME seller low willingness-to-pay and high subscription churn	Medium	High	Free tier lowers adoption barrier with no commitment; session history and CSV caching create switching cost over time; measurable time savings of 6+ hours per week demonstrated during onboarding to justify Starter and Pro subscription value

7. Implementation & Financials

7.1 Go-to-Market Strategy

AdKedai adopts a community-first, seller-partnership go-to-market strategy, launching in Kuala Lumpur as its beachhead market before expanding to Malaysia-wide coverage by Year 2 and broader SEA by Year 3. The strategy is structured in three phases.

Phase 1 — Community Seeding (Month 1–3): Partner with active SME seller communities on Facebook, Shopee seller forums, and TikTok creator networks to offer free Pro-tier access to 500 early adopters in

exchange for structured feedback and testimonials. Simultaneously engage MDEC's SME digitalisation programme to position AdKedai as a grant-eligible AI productivity tool.

Phase 2 — Paid Acquisition & Agency Channel (Month 4–12): Launch performance marketing campaigns targeting Shopee and TikTok Shop sellers in Malaysia. Onboard 10–20 SME-focused marketing agencies onto the Agency tier — each managing 5–30 seller accounts — transforming each agency into a distribution multiplier that eliminates individual seller customer acquisition costs.

Phase 3 — SEA Expansion & API Partnerships (Month 13–24): Localise seasonal event calendar for Indonesia, Thailand, and the Philippines; establish API licensing agreements with 2–3 platform-adjacent SaaS tools to embed AdKedai's recommendation engine natively within their interfaces.

7.2 Implementation Plan

Phase	Action	Timeline	Target
Product Hardening	Resolve API performance issues, complete CSV validation edge cases, expand automated test coverage to 95%+	Month 1–2	Production-ready platform; all CI/CD gates green
Community Beta Launch	500 early access users from KL SME seller communities; structured NPS and recommendation quality feedback loop; UI/UX iteration based on feedback	Month 2–4	500 active beta users; NPS ≥ 40
Public Paid Launch (Kuala Lumpur)	Launch all paid tiers publicly; begin performance marketing targeting KL Shopee and TikTok Shop sellers; onboard first KL agency partners.	Month 4–9	2,000 paying subscribers; RM 250K MRR
Agency Channel Scale	Onboard KL-based marketing agencies; launch white-label export and multi-	Month 9–15	6,000 paying subscribers; 20 agency partners

	account dashboard; API licensing negotiations		
Malaysia Expansion	Expand beyond KL to Selangor, Penang, Johor; localise recommendations for each city's seasonal selling patterns	Month 15–24	10,000+ paying subscribers across Malaysia
TOTAL	RM 5.2M investment	24 Months	10,000+ paying subscribers across Malaysia; 20+ agency partners

7.3 Revenue Projections (e.g 3-Year)

<i>Metric</i>	<i>Year 1</i>	<i>Year 2</i>	<i>Year 3</i>
Active Paying Subscribers	2,000	8,500	18,000
Average Revenue Per User (ARPU / month)	RM 125	RM 148	RM 162
Agency Partners	20	55	120
Subscription Revenue	RM 3.0M	RM 15.1M	RM 35.0M
API Licensing & Pay-Per-Session Revenue	RM 0.2M	RM 1.8M	RM 4.2M
<i>Total Revenue</i>	RM 3.2M	RM 16.9M	RM 39.2M
Total Operating Costs	RM 5.8M	RM 12.4M	RM 21.6M
<i>Net Position</i>	RM (2.6M)	RM +4.5M	RM +17.6M

Note: Year 1 figures reflect Kuala Lumpur-only operations. Year 2 reflects Malaysia-wide expansion. Year 3 reflects initial SEA market entry. Year 1 subscriber target revised to 2,000 to align with the KL SOM of 1,700–2,500 paying subscribers.

7.4 Key Financial Milestones

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- **Month 4:** 500 KL beta users completed feedback cycle; all paid tiers launched publicly; first paying subscribers acquired across Pay-Per-Session and Starter tiers.
- **Month 9:** 2,000 paying subscribers in KL; first API licensing agreement signed; monthly recurring revenue (MRR) exceeds RM 250,000.
- **Month 15:** 6,000 paying subscribers; 20 KL agency partners active; Malaysia expansion commenced to Selangor, Penang, and Johor; MRR exceeds RM 750,000.
- **Month 20:** Break-even achieved; positive contribution margin on all operations; cashflow-positive milestone reached.
- **Month 24:** 10,000+ paying subscribers across Malaysia; API licensing revenue established; SEA expansion planning initiated for Indonesia and Thailand.
- **Month 30:** Active subscriber base across Malaysia and initial SEA markets; Series A fundraising feasibility review initiated.